

Surprising innovations

There is some pretty impressive R&D going on in the world of 3D printing. Examples:

Living organs. Recently, a 3D elastic membrane has been developed using the 3D printing technology. This is a boon to the medical industry as it can help predict cardiac disorders and help the doctors correct them early. The specialty of this membrane is that it has tiny sensors printed on it, which provide detailed information even before the symptoms of the problem are exhibited by the patient physically.

Recent research has also seen the development of a 3D printed human liver and a heart, both made up of a transparent material and threaded with arteries and veins that are coloured. This is expected to help the physicians prevent the underlying complications during transplant.

3D printed edibles. It may sound crazy but it is a reality. A 3D printer called Foodini can create 3D-printed food with fresh ingredients, that too without the involvement of any cook. This printer can help create various shapes of the food that are generally difficult to create if you are not an expert. It can print various food items like chocolates, cookies of different shapes, hamburgers, etc. The only problem we see with this printer is that the food still needs to be cooked!

Green vehicles. Urbee is a 3D-printed car that is estimated to use about eight times less energy than an average car. It weighs less than 500kg and needs less than 38 litres of fuel to drive from San Francisco to New York. It marks an important landmark in the march towards green vehicles.

Ethical filament. Some 3D printing firms have partnered with the waste pickers in India to convert the waste plastic into filament for 3D printing. This would have both social as well as environmental impact on the society by empowering the waste pickers and eliminating the non-degradable plastic waste from the environment.

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at Boston, which can print carbon fibre. As mentioned earlier, though carbon fibre can be used to make things manually, it is tough to work with and time consuming as well. MarkOne can make working with carbon fibres easy and less time consuming, without the requirement to trim the object as only the required amount of carbon fibre is laid down. Thus a strong object can be created in less time with least effort. This printer not only works with carbon fibre but also has the ability to work with other types of composites like nylon, fiberglass and PLA (polylactic acid).

Then there is another printer called Duplicator 4, which is an assembled dual-extrusion 3D printer available in India. Although it is capable enough to be compared with MakerBot's Replicator 2x, it is almost one third in price. It can 3D print in two colours and also has the capability to use one extruder to print the actual object and the other to print the dissolvable material. It uses the FDM technology and can use both plastics and composites for the filament. Karan says, "Duplicator 4 is

a blatant copy, a clone or a duplicate of MakerBot design made in China and imported by some people in India."

Usually, the printers are not something you can assemble at home, but with 3D printers this is not the case. Besides buying a low-cost 3D

printer you also have the option to assemble your own 3D printer. There are many online global communities like RepRap that offer the do-it-yourself kits for assembling your own 3D printer.

Daniel Hutchison, co-founder, HYREL LLC says, "We have come up with Mk1 filament extruder and EMO-25 extruder, where EMO-25 can print precious-metal clay on all platforms and uses most of the emulsified filaments like PMC and silicon. "Biotz is introducing a disruptive portable 3D printer which would be affordable for each and every home by 2015," says Paul.

Want to try 3D printing?

New technologies require new tools, software and training to support it. Some companies like Redd Robotics, Cycloid System and Kurukshetra are organising workshops in India to help people understand the concept of 3D printing and to encourage this technology for the growth of the Indian market. Leading firm like Ultimatum and SAP Labs are working with ex-

perts in this industry to explore new opportunities for on-demand development of new technologies and software anywhere and anytime.

The software plays an important role in the designing and development before actual printing takes place. The 3D models designed are first verified and tested to ensure that the final product to be printed would meet the requirements of the customers. Karan says, "The major supplier of 3D printing software is Autodesk with their key software being AutoCAD. They are also releasing a number of free software related to 3D printing, one such example being 123D Design that is used for 3D modelling."

Then there are other software like Meshmixer that allows the designers to mix and match, stamp, script and paint one's own 3D printed designs. The user-friendly online tool like TinkerCAD allows the designers to convert their ideas into a CAD models for 3D printing. Harshad Karmarkar, research analyst, Patent iNSIGHT Pro says, "Although CAD/CAM software forms a base for 3D printing, companies have developed their own printers for specific applications. For example, 3D Systems has come up with a printer which prints on pizza."

Varying price

The main focus of the Indian market is low-priced 3D printers. Even though the consumers are apprehensive about quality, the main concern of the market is to ensure that the low-priced 3D printers meet the customers' requirement. Daniel Hutchison says, "The current price trends that we can see in the FDM or FFF market have been more towards making the cheapest printer possible for the consumer-level marketplace. SL has been price trending the same way although SLS is still an enigma at the moment." He opines, "It will be interesting to see how HP and other large companies would affect the 3D printing marketplace when they make their entrance in the near future." ●

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